

EE 446: TinyML

Lab 7: Wizard Magic Wand

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Today's Lab

- Both Software and Hardware Lab!
- Gathering data
- Training a model to recognize gesture
- Finetuning your model using your data
- Quantize the model for TinyML
- Deploy the model to the Arduino board



Lab 7 - Software - Road Map

1. Data Collection using Bluetooth
2. Labelled the data and Upload it to Google Drive
3. Rasterize the JSON files to Images
4. Train a CNN Model
5. Model Evaluation
6. Covert the TensorFlow model to TensorFlow Lite Micro model



What you need

- Mac, Linux or Windows laptop
- Arduino Nano BLE Sense 33 Rev2
- MicroUSB cable
- Possibly a USB C to USB A converter, if you're on a USB C laptop

[Optional] If you really really want to build magic wand ...

- A stick, about 12 inches (30 cm long)
- Sticky tape

Recording Gestures

1. Connect the board to the webpage using Bluetooth and Chrome's WebBLE API.
2. Moving the wand to collect the data.
3. The gestures are automatically split up by times when the wand is kept still.
4. Aim for at least 100 clean gestures of each type
5. Review and remove any sloppy or unrecognizable gestures using the trash can button
6. Label each gesture with its correct name (e.g., 0 or 1)
7. Download the reviewed and labeled data as a JSON text file for training purposes

Please refer to “***Lab 7 Data Collection Instruction.docx***”
provided for this lab in canvas under Files section

Training and Quantize the Model

Now Open [*EE446_TinyML_Lab7_\(Magic_Dataset_Training\).ipynb*](#)

(in Canvas → Files → Labs → Lab7)

in the tinymml-arduino python environment you have created for this class

Let's train our magic wand!

Lab 7 - Hardware - Road Map

1. Connect Arduino Boards to your computer
2. Open magic wand example
3. Update the model in the example and deploy the model
4. Open Serial Monitor and test the model



Deploy the Model

1. Use a USB cable to connect the Arduino Nano 33 BLE Sense to your machine. You should see the green LED power indicator come on when the board first receives power.
2. Open the magic_wand.ino sketch, which you can find in provided .zip file (Lab-7-Material.zip --> magic_wand_lab7.zip).
3. You'll then need to make two changes to the magic_wand.ino file to alert it of your number of gestures and gesture labels. These changes occur on lines 55-59, which currently read as:

```
//  
-----  
----- //  
  
// UPDATE THESE VARIABLES TO MATCH THE NUMBER AND LIST OF  
GESTURES IN YOUR DATASET //  
  
//  
-----  
----- //
```

Deploy the Model

```
constexpr int label_count = 10;  
  
const char * labels[label_count] =  
{"0", "1", "2", "3", "4", "5", "6", "7", "8", "9"};
```

- a. Update the label_count to reflect the number of gestures in your dataset.
- b. Update the list of labels to reflect the gestures in your dataset. Note: the order matters! Make sure it matches the alphanumeric order as printed out in the training script!

Deploy the Model

4. You'll also need to update the model data as we've done with the previous examples. The model data can be found in the `magic_wand_model_data.cpp` file. As we've done in the past, make sure to only update the binary values and leave the cpp syntax constant.

5. When you save, just like with the KWS examples, you will be asked to save a copy. Again, we suggest that you make a folder called e.g., TinyML inside of your Arduino folder. You can find your main Arduino folder either inside of your Documents folder or in your Home folder, and save it in that folder with a descriptive name like `magic_wand_custom`. That said, you can save it wherever you like with whatever name you want!

Deploy the Model

6. As always, use the Tools drop-down menu to select the appropriate Port and Board, and use the rightward arrow to upload / flash the code.
7. Now, open the serial monitor and test out your custom model. As a reminder, the serial monitor will output first ASCII art showing the gesture you just performed and below it will be the best match label as well as a confidence score between -128 and 127. The confidence score indicates how strongly the model believes you performed the gesture. Do note that every time you move the board and then stop, a new gesture will be processed, so don't be surprised to get some odd results as you move the board to prepare for a gesture.



What you Should Submit

You are required to complete and submit the following four tasks. All Lab 7 files must be uploaded to the GitHub repository that you are maintaining for this class. Then submit the GitHub repository link, or the direct Lab 7 folder link, through the Canvas submission page for Lab 7.

✓ Task 1: Deploy Baseline int8 Quantized Model

- Deploy the **int8 quantized model *before* fine-tuning** onto your Arduino Nano 33 BLE Sense.
 - Draw each digit (0–9) and capture screenshots of the **Serial Monitor** output.
 - Each screenshot **must clearly show**:
 - The ASCII representation of the digit you drew.
 - The predicted digit.
 - The prediction confidence value (score).
-

✓ Task 2: Collect Personal Gesture Data

- Follow the detailed steps in the **Lab 7 Data Collection Instruction.docx** to collect your own digit gesture data using the Magic Wand interface.
 - Generate your `wanddata.json` file and use it in the notebook for fine-tuning.
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What you Should Submit

✓ Task 3: Deploy Fine-Tuned int8 Quantized Model

- Fine-tune your model using your collected data.
 - Quantize the fine-tuned model using **int8 quantization**, and deploy it on your Arduino Nano 33 BLE Sense.
 - Again, draw each digit (0–9) and take **screenshots of the Serial Monitor output**.
 - Each screenshot **must clearly show**:
 - The ASCII representation of the digit.
 - The predicted digit.
 - The prediction confidence value (score).
-

✓ Task 4: Compare Results and Reflect

- Compare the performance of the **baseline model (Task 1)** vs the **fine-tuned model (Task 3)**.
 - Briefly explain:
 - What differences did you observe?
 - Which model performed better and why?
 - Did fine-tuning help for your specific gesture style?
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What you Should Submit

Upload all Lab 7 materials to the GitHub repository that you are maintaining for this class. Your GitHub submission should include all relevant files needed to reproduce and review your work, including:

- Your completed Jupyter notebook.
- Your collected `wanddata.json` file.
- The Arduino `.ino` file used for the **baseline int8 quantized model before fine-tuning**.
- The Arduino `.ino` file used for the **fine-tuned int8 quantized model**.
- Any generated model files, converted model files, header files, or supporting files used for deployment.
- Screenshots of the Serial Monitor outputs for digits 0–9 before fine-tuning.
- Screenshots of the Serial Monitor outputs for digits 0–9 after fine-tuning.
- A short reflection comparing the baseline and fine-tuned models. This reflection can be included either in the notebook or as a separate PDF/Markdown file.

Submit the GitHub repository link, or the direct link to your Lab 7 folder inside the repository, through the **Lab 7 submission page on Canvas**.

💡 *Tip:* Be sure to perform inference for all 10 digits (0–9) in both Task 1 and Task 3. Also make sure your GitHub link is accessible to the teaching team.

